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Synchronization of Rossler chaotic systems via hybrid adaptive backstepping/sliding mode control

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ABSTRACT

In this paper, a hybrid control approach combining adaptive backstepping and fuzzy sliding mode control is proposed for the synchronization of two Rossler chaotic systems. The proposed approach provides fast, high accuracy chaos synchronization in presence of unknown system parameters and external disturbances. The adaptive backstepping control law is developed to stabilize the synchronization error dynamics arising from the mismatch between the drive and response systems. Both unknown system parameters and unknown external disturbance are estimated by the adaptive backstepping control approach in a way that the overall system with the controller becomes stable. As a result, the requirement to have *a priori* knowledge about the unknown parameters, e.g., disturbance bounds, is nullified. The fuzzy sliding mode controller is designed to accompany the adaptive backstepping to confine the synchronization error and enhance the synchronization speed between the two chaotic systems. Performance of the proposed approach is investigated for a numerical example and compared against commonly used chaos synchronization techniques including optimal control, active control, and adaptive backstepping control method. Simulation results indicate the superior performance of the hybrid control method over state-of-the-art for the synchronization of Rossler chaotic systems.

1. Introduction

Chaos theory has been studied in many application areas, including chemical [1,2], ecological [3,4] economical [5], and communication systems [6,7]. The work of Pecora et al. [8] was among the pioneer discoveries of chaos synchronization, in which two chaotic systems, namely drive and response systems, were synchronized. Chaos synchronization has received great attention in recent years mainly due to its application in secure broadband communication and control processing [9]. For instance, it plays an important role in secure communication [10,11], where the confidential information such as financial institutions account or government information needs to be transmitted in a secure matter in presence of internal and external unknowns.

Chaos synchronization entails the process of oscillating two chaotic systems in a synchronized manner. To this end, a proper control algorithm is needed to stabilize the synchronization error dynamics that arise from the mismatch between the drive and response systems to an equilibrium point such as the origin. As a result, all states of the response system will have a parallel behavior to those of the corresponding drive system [10]. Various control methods, including linear feedback control [12–14], nonlinear control [15–21], optimal control [22–26], active control [27–29], and adaptive control [30–33], have been proposed for the synchronization of different chaotic systems. In [14], the synchronization of two identical chaotic electronic circuits with a linear feedback controller was studied, and the range of controller parameters was estimated by a Lyapunov exponent approach. In [15], a nonlinear approach using sliding mode control was proposed for the synchronization of two identical Pehlivan chaotic systems [34]. Synchronization of non-identical fractional order chaotic systems using sliding mode control was considered in [21]. Adaptive sliding

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mode control was employed in [35] for synchronization of communication systems in wireless sensors. In [36], the combination of nonlinear feedback and sliding mode control was used for synchronization of chaotic systems with time delays. In [22], an optimal control approach based on the Lyapunov–Bellman technique [37] was presented to derive control laws in synchronization of Rossler chaotic systems. Most of these approaches presume the parameter values of the chaotic systems are known, which may not be a practical assumption due to inherent uncertainties in the value of parameters in chaotic systems [27].

Active and adaptive control methods, on the other hand, are designed based on the unknown system parameters. In [28], a Lyapunov-based active control approach was proposed to synchronize Lorenz [38] and Chen [39] chaotic systems. In [40], an adaptive control approach was used to synchronize Rossler chaotic systems with unknown parameters. Adaptive backstepping approaches were also used for the synchronization of Arneodo chaotic systems [41] and Rosller oscillators [42]. One drawback of these approaches is their inferior performance in terms of fast synchronization of chaotic systems. As demonstrated in the literature [43], they fail to synchronize the two chaotic systems in a short amount of time, which is highly demanded in secure communication and other application areas to accurately and safely receive and preserve data [6,7]. Furthermore, there might exist external sources of uncertainties such as unknown disturbances that can potentially affect the performance of chaos synchronization systems. To the best of Authors' knowledge, the effect of external disturbances on the chaos synchronization was not previously studied in the literature. Thus, there is a remarkable gap in the literature with respect to an efficient control approach that can provide a quick synchronization between the two chaotic systems in presence of both internal parametric uncertainties and external unknown disturbances.

This work intends to bridge the gap by proposing a hybrid control approach that combines the adaptive backstepping with a novel fuzzy sliding mode control to provide fast, high accuracy synchronization of chaotic systems in presence of unknown system parameters and external disturbances. In the first step, an adaptive backstepping control law was developed to stabilize the synchronization error dynamics arising from the mismatch between the drive and response systems. Unknown system parameters along with the unknown external disturbance were estimated by the adaptive backstepping control approach in such a way that the overall system with the controller would become asymptotically stable. As a result, the requirement to have a priori knowledge about the unknown disturbance bounds was nullified. The adaptive backstepping control has a simple recursive structure with only one control law to stabilize all the error dynamics of the system. In the second step, a fuzzy sliding mode controller was designed to accompany the adaptive backstepping controller in order to accelerate the synchronization between the drive and response systems and prevent large deviations of error dynamics. The sliding mode design parameters and controller gains were tuned with two separate fuzzy systems. The stability of the overall system with the hybrid adaptive backstepping/sliding mode controller was studied through dissipativity theorem.

The rest of the paper is organized as follows: Section 2 describes the Rossler chaotic systems. The proposed hybrid control solution is described in Section 3. Stability analysis of the overall system with the hybrid controller is presented in Section 4. A numerical example and comparison against the state-of-the-art are explained in Section 5. Conclusions are drawn in Section 6.

2. Rossler system

The Rossler chaotic system is described as

$$\begin{aligned} \dot{x} &= -y - z \\ \dot{y} &= x + ay \\ \dot{z} &= b + z(x - c) \end{aligned} \tag{1}$$

where a, b, and c are three constant parameters. Assuming $x_1 = y$, $x_2 = x$, and $x_3 = z$, the Rossler model for drive and response systems can be written respectively as (2) and (3).

$$\begin{aligned} \dot{x}_1 &= ax_1 + x_2 \\ \dot{x}_2 &= -x_1 - x_3 \\ \dot{x}_3 &= b + x_3(x_2 - c) \end{aligned} \tag{2}$$

$$\begin{aligned} \dot{y}_1 &= \hat{a}y_1 + y_2 \\ \dot{y}_2 &= -y_1 - y_3 \\ \dot{y}_3 &= \hat{b} + y_3(y_2 - \hat{c}) + \hat{d} \end{aligned} \tag{3}$$

where x_i and y_i, $i = 1, 2, 3$ represent states dynamics for the drive and response system, respectively. Also, $\hat{a}$, $\hat{b}$ and $\hat{c}$ are unknown system parameters and $\hat{d}$ is unknown external disturbance applied to the response system. The error dynamics system is obtained as

$$\begin{aligned} \dot{e}_1 &= e_2 + \hat{a}y_1 - ax_1 \\ \dot{e}_2 &= -e_1 - e_3 \\ \dot{e}_3 &= (\hat{b} - b) + y_2y_3 - x_2x_3 - \hat{c}y_3 + cx_3 + \hat{d} \end{aligned} \tag{4}$$

where $e_i = y_i - x_i$, $i = 1, 2, 3$. The goal is to design a controller for coupling drive and response systems such that the global synchronization holds. It implies that the synchronization error quickly converges to the origin and remain in its vicinity:

$$\lim_{t\to\infty} \|e(t)\| = 0 \tag{5}$$

where $\|\|$ indicates the Euclidean norm.

3. Hybrid adaptive backstepping/sliding mode control

3.1. Adaptive backstepping

Originally developed by Peter Kokotovic in the early 1990's [44], backstepping is a recursive control technique that relates a control Lyapunov function with the design of a feedback controller to stabilize a class of nonlinear dynamical systems. In this work, an adaptive backstepping control approach was designed to drive the error dynamics system of (4) to the origin. The unknown parameters $\hat{a}$, $\hat{b}$ and $\hat{c}$ along with the unknown disturbance $\hat{d}$ were also adaptively estimated in the proposed method. The controller u_b was considered in the response system. Thus, the response system dynamics with the adaptive backstepping controller are shown as

$$\begin{aligned} \dot{y}_1 &= \hat{a}y_1 + y_2 \\ \dot{y}_2 &= -y_1 - y_3 \\ \dot{y}_3 &= \hat{b} + y_3(y_2 - \hat{c}) + u_b + \hat{d} \end{aligned} \tag{6}$$

In the backstepping approach, one control law is sufficient to stabilize the system. With the addition of the controller, the error dynamics system – difference between drive and response system – is found as

$$\begin{aligned} \dot{e}_1 &= e_2 + \hat{a}y_1 - ax_1 \\ \dot{e}_2 &= -e_1 - e_3 \\ \dot{e}_3 &= (\hat{b} - b) + y_2y_3 - x_2x_3 - \hat{c}y_3 + cx_3 + u_b + \hat{d} \end{aligned} \tag{7}$$

Theorem 3.1. *The error dynamics system of (7) is asymptotically stable in the vicinity of origin if the control law u_b is designed as*

$$u_b = -y_2y_3 + x_2x_3 - cx_3 + k_2\dot{z}_2 - k_3z_3 + z_2 + 2\dot{\hat{a}}\dot{e}_2 + \ddot{\hat{a}}e_1 + k_1\ddot{z}_1 + \dot{z}_1 + \hat{a}\ddot{e}_1 + \hat{d} \tag{8}$$

where k_1, k_2 and k_3 are constant parameters and z_1, z_2, and z_3 are defined as

$$\begin{aligned} z_1 &= e_1 \\ z_2 &= e_2 - \phi_1,\ \phi_1 = -\hat{a}e_1 - k_1z_1 \\ z_3 &= e_3 - \phi_2,\ \phi_2 = k_2z_2 + \dot{\hat{a}}e_1 + z_1 + k_1\dot{z}_1 + \hat{a}\dot{e}_1 \end{aligned} \tag{9}$$

Proof. (1) Let $z_1 = e_1$

By using $\dot{e}_1$ from (7), the derivative of z_1 is found as

$$\dot{z}_1 = \dot{e}_1 = e_2 + \hat{a}y_1 - ax_1 \tag{10}$$

(2) Let $z_2 = e_2 - \phi_1 \Rightarrow e_2 = z_2 + \phi_1$
which can be substituted in (10) as

$$\dot{z}_1 = z_2 + \phi_1 + \hat{a}y_1 - ax_1 \tag{11}$$

The first Lyapunov function is considered as

$$v_1(t) = \frac{1}{2}z_1^2 \tag{12}$$

The derivative of $v_1(t)$ is

$$\dot{v}_1(t) = z_1\dot{z}_1 = z_1(z_2 + \phi_1 + \hat{a}y_1 - ax_1) \tag{13}$$

Manipulating (13) yields

$$\dot{v}_1(t) = z_1(z_2 + \phi_1 + \hat{a}y_1 - ax_1) = z_1(z_2 + \phi_1 + \hat{a}y_1 - ax_1 + \hat{a}x_1 - \hat{a}x_1) = z_1(z_2 + \phi_1 + \hat{a}e_1 + (\hat{a} - a)x_1) \tag{14}$$

Without loss of generality, (14) can be rewritten as

$$\dot{v}_1(t) = z_1(z_2 + \phi_1 + \hat{a}e_1 + (\hat{a} - a)x_1 + k_1z_1 - k_1z_1) \tag{15}$$

where k_1 is a constant parameter. By substituting ϕ_1 from (9) into (15), $\dot{v}_1(t)$ can be written as

$$\begin{aligned}\dot{v}_1(t) &= z_1(z_2 - \hat{a}e_1 - k_1 z_1 + \hat{a}e_1 + (\hat{a}-a)x_1 + k_1 z_1 - k_1 z_1)\\ &= z_1 z_2 + (\hat{a}-a)x_1 z_1 - k_1 {z_1}^2 = z_1 z_2 + (\hat{a}-a)x_1 e_1 - k_1 {z_1}^2\end{aligned} \tag{16}$$

(3) Let $z_3 = e_3 - \phi_2 \Rightarrow e_3 = z_3 + \phi_2$
The Lyapunov function $v_2(t)$ is considered as

$$v_2(t) = \frac{1}{2}z_2^2 + v_1 + \frac{1}{2}(\hat{a}-a)^2 \tag{17}$$

Derivative of (17) can be found as

$$\begin{aligned}\dot{v}_2(t) &= z_2\dot{z}_2 + \dot{v}_1 + \dot{\hat{a}}(\hat{a}-a)\\ &= z_2(\dot{e}_2 - \dot{\phi}_1) + z_1 z_2 + (\hat{a}-a)x_1 e_1 - k_1 {z_1}^2 + \dot{\hat{a}}(\hat{a}-a)\\ &= z_1 z_2 + (\hat{a}-a)x_1 e_1 - k_1 z_1^2 + z_2(-e_1 - e_3 + \hat{a}\dot{e}_1 + \dot{\hat{a}}e_1 + k_1\dot{z}_1) + \dot{\hat{a}}(\hat{a}-a)\end{aligned} \tag{18}$$

By substituting $e_3 = z_3 + \phi_2$ from (9) into (18), $\dot{v}_2(t)$ can be represented as

$$\dot{v}_2(t) = (\hat{a}-a)x_1 e_1 - k_1 z_1^2 + z_2(-z_3 - \phi_2 + \hat{a}\dot{e}_1 + k_2 z_2 - k_2 z_2 + \dot{\hat{a}}e_1 + k_1\dot{z}_1 + z_1) + \dot{\hat{a}}(\hat{a}-a) \tag{19}$$

Substituting $\phi_2 = k_2 z_2 + \dot{\hat{a}}e_1 + k_1\dot{z}_1 + z_1 + \hat{a}\dot{e}_1$ from (9) and assuming the dynamics of estimated $\hat{a}$ as $\dot{\hat{a}} = -x_1 e_1$, (19) can be updated as

$$\dot{v}_2(t) = -k_1 z_1^2 - k_2 z_2^2 - z_2 z_3 \tag{20}$$

Now, considering the Lyapunov function $v_3(t)$ as

$$v_3(t) = \frac{1}{2}z_3^2 + v_2 + \frac{1}{2}\hat{c}^2 + \frac{1}{2}(\hat{b}-b)^2 + \frac{1}{2}\hat{d}^2 \tag{21}$$

its derivative is calculated as

$$\dot{v}_3(t) = z_3\dot{z}_3 + \dot{v}_2 + \dot{\hat{c}}\hat{c} + \dot{\hat{b}}(\hat{b}-b) + \dot{\hat{d}}\hat{d} = z_3\dot{z}_3 - k_1 z_1^2 - z_2 z_3 + \dot{\hat{b}}(\hat{b}-b) + \dot{\hat{c}}\hat{c} + \dot{\hat{d}}\hat{d} \tag{22}$$

Taking the derivative of $z_3 = e_3 - \phi_2$ yields

$$\dot{z}_3 = \dot{e}_3 - \dot{\phi}_2 \tag{23}$$

The derivative of ϕ_2 can be obtained from (9) as

$$\dot{\phi}_2 = k_2\dot{z}_2 + \dot{\hat{a}}\dot{e}_1 + \ddot{\hat{a}}e_1 + k_1\ddot{z}_1 + \dot{z}_1 + \dot{\hat{a}}\dot{e}_1 + \hat{a}\ddot{e}_1 \tag{24}$$

Substituting (23) and (24) in (22) yields

$$\begin{aligned}\dot{v}_3(t) &= -k_1 z_1^2 - z_2 z_3 + z_3\big((\hat{b}-b) + y_2 y_3 - x_2 x_3 - \hat{c}y_3 + \mathrm{c}x_3 + u_b + \hat{d} - k_2\dot{z}_2\\ &\quad -\dot{\hat{a}}\dot{e}_1 - \ddot{\hat{a}}e_1 - k_1\ddot{z}_1 - \dot{z}_1 - \dot{\hat{a}}\dot{e}_1 - \hat{a}\ddot{e}_1\big) + \dot{\hat{b}}(\hat{b}-b) + \dot{\hat{c}}\hat{c} + \dot{\hat{d}}\hat{d}\\ &= -k_1 z_1^2 - z_2 z_3 + z_3(y_2 y_3 - x_2 x_3 + cx_3 + u_b - \hat{d} - k_2\dot{z}_2 + k_3 z_3 - k_3 z_3 + z_2 - z_2 - 2\dot{\hat{a}}\dot{e}_1\\ &\quad -\ddot{\hat{a}}e_1 - k_1\ddot{z}_1 - \dot{z}_1 - \hat{a}\ddot{e}_1) + (\hat{b}-b)(\dot{\hat{b}} + z_3) + \hat{d}(\dot{\hat{d}} + 2z_3) + \hat{c}(\dot{\hat{c}} - y_3 z_3)\end{aligned} \tag{25}$$

Consequently, estimation dynamics of b, c and d is assumed as

$$\begin{aligned}\dot{\hat{b}} &= -z_3\\ \dot{\hat{c}} &= y_3 z_3\\ \dot{\hat{d}} &= -2z_3\end{aligned} \tag{26}$$

and control law is designed as

$$u_b = -y_2 y_3 + x_2 x_3 - cx_3 + k_2\dot{z}_2 - k_3 z_3 + z_2 + 2\dot{\hat{a}}\dot{e}_2 + \ddot{\hat{a}}e_1 + k_1\ddot{z}_1 + \dot{z}_1 + \hat{a}\ddot{e}_1 + \hat{d} \tag{27}$$

and $\dot{v}_3(t)$ is obtained as

$$\dot{v}_3(t) = -k_1 z_1^2 - k_2 z_2^2 - k_3 z_3^2 \tag{28}$$

and is negative definite. Thus, the dynamics of z_is' system are globally asymptotically stable in the equilibrium point of origin, according to the Lasalle theorem [45]. As a result, the dynamics of error system is asymptotically stabilized and $\hat{a}$, $\hat{b}$, $\hat{c}$ and $\hat{d}$ are restricted in the neighborhood of desired values.

It should be noted that there is no meaningful relationship between the values of estimated parameters $\hat{a}$, $\hat{b}$, $\hat{c}$, $\hat{d}$, nor between their derivatives $\dot{\hat{a}}$, $\dot{\hat{b}}$, $\dot{\hat{c}}$, $\dot{\hat{d}}$. Also, the proposed backstepping controller is not limited to the Rossler systems. As a result, when the chaotic system model changes, e.g., Lorenz, Lu, or Chen system, the equations for estimating unknown parameters and disturbance will vary to meet the Lyapunov stability conditions.

Table 1
Fuzzy rules for δ_i calculation.

S_i	NB	NM	Z	PM	PB
δ_i	NB	NM	Z	PM	PB

NB = Negative Big, NM = Negative Medium, Z = Zero, PM = Positive Medium, PB = Positive Big

3.2. Fuzzy sliding mode control

To confine the synchronization error within a specific range and enhance the synchronization speed between the two chaotic systems, a fuzzy sliding mode controller was designed to accommodate the adaptive backstepping controller. The resulting hybrid control approach has a notably higher performance than a pure adaptive backstepping controller in terms of both synchronization speed and error, as will be later demonstrated in Section 5.

The dynamics of response system with the hybrid controller consisting of both adaptive backstepping and sliding mode control laws are derived as

$$\begin{aligned} \dot{y}_1 &= \hat{a}y_1 + y_2 + \beta_1 u_{s_1} \\ \dot{y}_2 &= -y_1 - y_3 + \beta_2 u_{s_2} \\ \dot{y}_3 &= \hat{b} + y_3(y_2 - \hat{c}) + u_b + \beta_3 u_{s_3} + \hat{d} \end{aligned} \tag{29}$$

where u_b and u_{si}, $i = 1,2,3$ denote control laws for adaptive backstepping and sliding mode controllers, respectively. Also, β_i, $i = 1,2,3$ are the gains of sliding mode controller.

The error dynamics system is derived by subtracting (2) from (29) as

$$\begin{aligned} \dot{e}_1 &= e_2 + \hat{a}y_1 - ax_1 + \beta_1 u_{s_1} \\ \dot{e}_2 &= -e_1 - e_3 + \beta_2 u_{s_2} \\ \dot{e}_3 &= (\hat{b} - b) + y_2 y_3 - x_2 x_3 - \hat{c}y_3 + cx_3 + u_b + \beta_3 u_{s_3} + \hat{d} \end{aligned} \tag{30}$$

In the sliding mode control, a weighted summation of error and higher order derivatives of error constitute a dynamic called sliding surface, i.e.,

$$\begin{aligned} &S_i(t) = (\frac{d}{dt} + \lambda_i)^{n-1} e_i \overset{n=2}{\Rightarrow} S_i(t) = \dot{e}_i + \lambda_i e_i \\ &(\lim_{t\to\infty} S(t) = 0 \text{ and } \lim_{t\to\infty} \dot{S}(t) = 0) \end{aligned} \tag{31}$$

where S_i, $i = 1,2,3$ are the sliding surfaces, λ_i is a positive constant, and $e_i, i = 1,2,3$ are the result of subtracting the desired state (origin) from the state dynamics. The aim is to derive a control law by which the error is directed towards origin. The sliding condition is used to establish an appropriate condition as

$$\begin{aligned} &\frac{d}{dt}(S^T S) \le -\eta\,|S| \\ &(S = \begin{bmatrix} S_1 & S_2 & S_3 \end{bmatrix}^T) \\ &\Rightarrow S^T \dot{S} \le -\eta\,|S(t)| \\ &\Rightarrow \ddot{e}_i + \lambda_i \dot{e}_i \le -\eta_i \mathrm{sgn}(S_i) \end{aligned} \tag{32}$$

where η_i, $i = 1,2,3$ are positive constants.

To have an asymptotically stable system, and hence a robust behavior, the sliding mode control laws u_{si}, $i = 1,2,3$ are designed to direct system dynamics to the sliding surfaces ($S(t) = 0$) [21]. Given the sliding condition of (32), the sliding mode control laws are obtained as

$$\begin{aligned} &u_{s_i} = S_i - \delta_i \mathrm{sgn}(S_i) \\ &\delta_i \ge \left| F - \hat{F} \right| \end{aligned} \tag{33}$$

where sgn is the sign function and δ_i, $i = 1,2,3$ are positive constants. Also F and $\hat{F}$ are uncertain dynamics and estimation of uncertain dynamics, respectively. To cancel the chattering phenomenon [46,47], the sign function is replaced with a saturation function to update the control law as

$$u_{s_i} = S_i - \delta_i\,\mathrm{sat}(S_i) \tag{34}$$

Since the bound of system uncertainty is not available , δ_is need to be estimated. To this end, a novel estimator was designed using Mamdani fuzzy system with five fuzzy membership functions ($M = 5$) for both estimator input(S_i) and output δ_i. The fuzzy rules for calculating δ_is are demonstrated in Table 1.

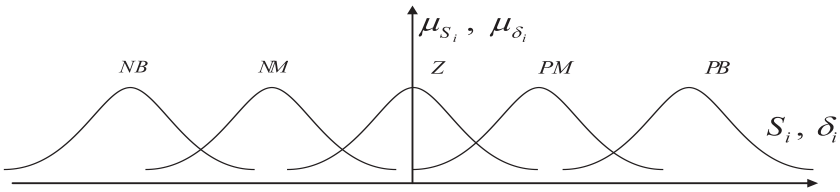

Fig. 1. Membership functions for S_i and δ_i.

Table 2
Fuzzy rules for tuning β_i.

e_i	NB	NM	Z	PM	PB
β_i	B	M	Z	M	B

NB = Negative Big, NM = Negative Medium, Z = Zero, PM = Positive Medium, PB = Positive Big, B = Big, M = Medium, Z = Zero

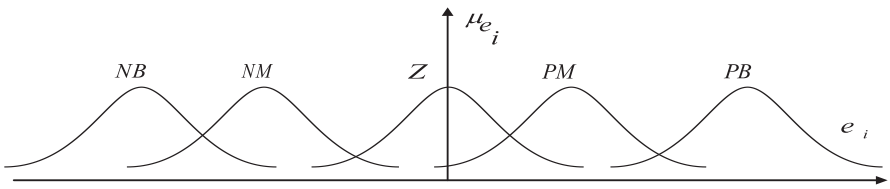

Fig. 2. Membership functions for e_i.

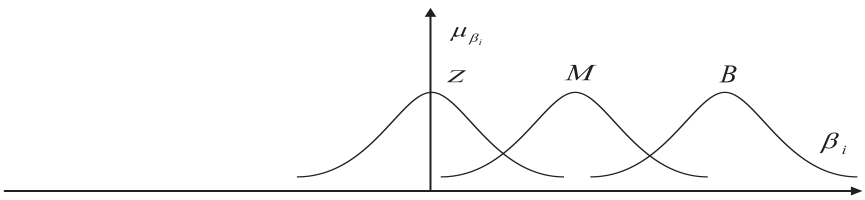

Fig. 3. Membership functions for β_i.

Five Gaussian membership functions for S_i and δ_i are depicted in Fig. 1. Also, the singleton fuzzifier, product inference engine, and the center of average defuzzfier were used. δ_is were calculated as

$$\delta_i = \frac{\sum_{l=1}^{M} \overline{p}_i^l \mu_{S_i^l}}{\sum_{l=1}^{M} \mu_{S_i^l}}, \quad i = 1, 2, 3 \tag{35}$$

where $\overline{p}_i^l$ were the center of membership functions for δ_i and μ_{s_i} were the membership functions for S_i.

To properly apply the sliding mode control laws to the system, the controller gains β_is were tuned to limit the synchronization error within $[|e_{threshold}|, +\infty]$ where $|e_{threshold}|$ is the absolute value of a predetermined synchronization error threshold. As a result, the sliding mode controller affects the system dynamics more considerably when the synchronization error is within the aforementioned range. The controller gains β_is were tuned using another Mamdani fuzzy approach. The fuzzy rules for tuning β_is are shown in Table 2. Sliding mode control coefficients (β_is) are adjusted on the basis of rules presented in Table 2.

Five Gaussian membership functions for e_i (Fig. 2) and three Gaussian membership functions for β_i (Fig. 3) were used.

The singleton fuzzifier, product inference engine, and the center of average defuzzfier were applied to compute sliding mode controller gains as

$$\beta_i = \frac{\sum_{l=1}^{N} \overline{y}_i^l \mu_{e_i^l}}{\sum_{l=1}^{N} \mu_{e_i^l}}, \quad i = 1, 2, 3 \tag{36}$$

where $\overline{y}_i^l$ were the center of β_i membership functions and μ_{e_i} were e_i membership functions.

Fig. 4 demonstrates the overall configuration of the chaotic systems along with the adaptive backstepping and fuzzy sliding mode controller. While the proposed adaptive backstepping control approach stabilizes the system around the origin, as proven in Section 3.1, its performance will be notably enhanced by the sliding mode controller that compensates for the error deviations from the origin. In other words, the sliding mode controller was designed to accompany the adaptive backstepping controller in order to

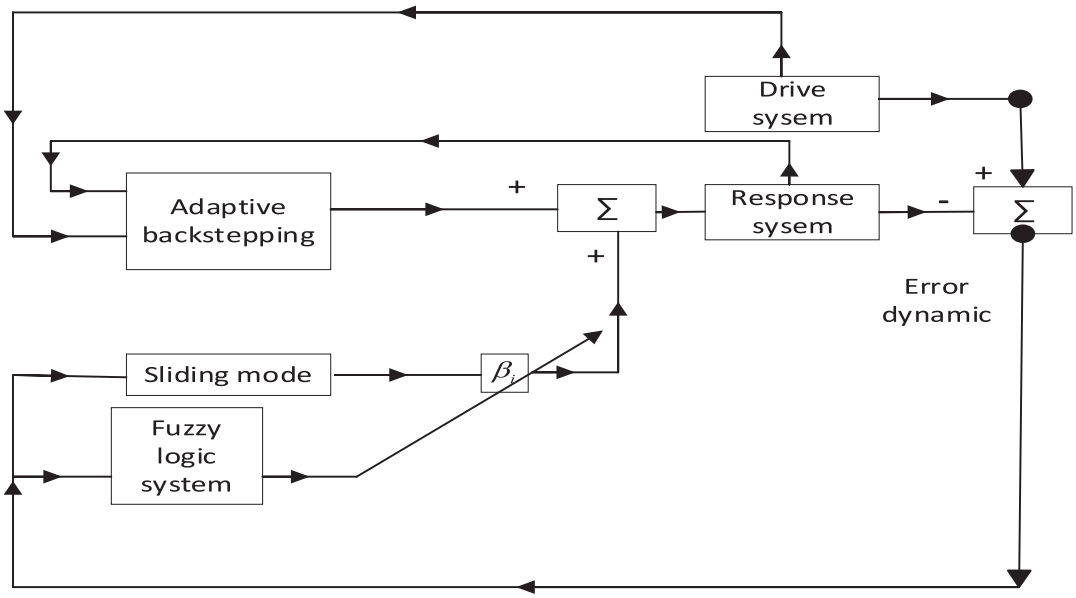

Fig. 4. Overall configuration of the drive and response system with the hybrid adaptive backstepping/fuzzy sliding mode controller.

accelerate the synchronization between the drive and response systems and prevent large deviations of error dynamics. Accelerated chaos synchronization leads to higher performance in application such as secure communication. Also, confining the error dynamics in the vicinity of origin helps with the accuracy of synchronization between the drive and response systems.

4. System stability

The stability of the overall system – with this hybrid controller – was studied using the dissipativity theory [48,49], which is independent from the system dynamics. Through dissipativity theory, input/output properties from a state space model is derived to establish bounds on the state trajectories using bounds on the storage function [49] (see Appendix A for further discussion).

Definition 1. The supply rate function J, which is integrable and independent of the states, is dissipative, and hence the system is stable, if there exists a continuously differentiable storage function $V \geq 0$ such that

$$\begin{aligned} &V(e(t)) - V(e(0)) \leq \int_0^T J[u_t(\mathrm{t}), e(t)]dt \\ &\Rightarrow V(e(t)) - V(e(0)) - \int_0^T J[u_t(\mathrm{t}), e(t)]dt \leq 0, \quad \int_0^T J[u_t(\mathrm{t}), e(t)]dt \leq +\infty, V(e(0)) = 0 \end{aligned} \tag{37}$$

where $u_t(t) = [\beta_1(t)u_{s_1}(t) \quad \beta_2(t)u_{s_2}(t) \quad \beta_3(t)u_{s_3}(t) + u_b(\mathrm{t})]^T$ is the total control system vector as the input and $e(t) = [e_1(t) \quad e_2(t) \quad e_3(t)]^T$ is the synchronization error vector as the output. The nominated supply rate function J and the corresponding storage function V are considered as

$$\begin{aligned} &J[u_t(t), e(t)] = \gamma_1 e^T(t)e(t) + \gamma_2 {u_t}^T(t)u_t(t) \\ &V(e(t)) = \frac{1}{2}e^T(t)Qe(t) \end{aligned} \tag{38}$$

where γ_1 and γ_2 are constant values and Q is a constant matrix. Substituting (38) in (37) yields

$$\frac{1}{2}e^T(t)Qe(t) - \int_0^T [\gamma_1 e^T(t)e(t) + \gamma_2 {u_t}^T(t)u_t(t)]dt \leq 0 \tag{39}$$

The inequality condition of (39) must be enforced to satisfy the stability of the overall system. To elucidate further, if the nominated supply rate function J is negative definite for $u_t(t) = 0$, the dynamics' trajectory is bounded in a subset of the dynamic space (see Appendix A, Special Case). Theorem 4.1 substantiates the stability of a dissipative dynamic system:

Theorem 4.1. *An equilibrium point $x^* \in X$ of a dissipative dynamic system Γ is stable if the storage function V is continuous and attains a strong local minimum at x^*. Moreover V is a Lyapunov function in the neighborhood of x^* [50].*

Proof. It suffices to show that V is monotone and non-increasing at $t = t_0$, if $\|x^* - x_0\|$ is sufficiently small. Given the dissipation inequality of (37), V is indeed monotone and non-increasing for all $t \geq t_0$ and for all x^*.

5. Numerical example

In this Section, the hybrid adaptive backstepping/sliding mode controller was applied to synchronize two Rossler chaotic systems with the nominal value of parameters. Synchronization performance of the proposed hybrid controller was compared against three other chaos synchronization techniques: Optimal control [22], active control [27,28], and pure adaptive backstepping [42]. The optimal control approach [22] uses the Lyapunov–Bellman technique to derive control laws for synchronization of chaotic systems in presence of unknown system parameters. Chaos synchronization can be achieved in both finite and infinite time interval using optimal feedback control approach [22]. Active control [27,28] strategy is a Lyapunov-based method in which the control law is proposed beforehand in a way that the Lyapunov condition holds (see Appendix B for the derivation of active control law). The popularity of active control is mainly due to its simplicity and flexibility, compared with other synchronization methods [51]. Adaptive backstepping [42], another popular synchronization technique, was selected to be compared against the proposed hybrid adaptive backstepping/sliding model control to investigate the effectiveness of fuzzy sliding mode controller in the hybrid control approach. The derivation of adaptive backstepping control law for synchronization of Rossler chaotic systems in presence of unknown system parameters and external disturbances was described in Section 3.1. Performance comparison was conducted with a statistical test.

5.1. Results

Consider the Rossler system of (1) with nominal parameters $a = 0.2$, $b = 0.2$, $c = 5.7$ and initial values of, $x_i(0) = y_i(0) = 0$ $(i = 1, 2, 3)$. The system has chaotic behavior as shown in [43]. The synchronization of drive and response systems with the hybrid adaptive backstepping/sliding mode controller was considered. The nominal values for the controller design parameters were chosen as

$$\begin{aligned} &k_1 = k_2 = k_3 = 1 \\ &\lambda_1 = \lambda_2 = \lambda_3 = 1 \end{aligned} \tag{40}$$

For δ_i calculation, the first fuzzy system parameters were chosen as

$$\begin{aligned} &\text{membership functions variance}: \quad S_1 = 3, S_2 = 2, S_3 = 0.75 \\ &\text{membership functions variance}: \quad \delta_1 = 0.25, \delta_2 = 0.25, \delta_3 = 0.25 \\ &S_{1\,\min} = -12, S_{1\,\max} = 12, S_{2\,\min} = -8, S_{2\,\max} = 8, S_{3\,\min} = -3, S_{3\,\max} = 3 \\ &\delta_{1\,\min} = -1, \delta_{1\,\max} = 1, \delta_{2\,\min} = -1, \delta_{2\,\max} = 1, \delta_{3\,\min} = -1, \delta_{3\,\max} = 1 \end{aligned} \tag{41}$$

Also, the second fuzzy system parameters for β_i calculation were set as

$$\begin{aligned} &\text{membership function variance}: \quad e_1 = 2, e_2 = 1.25, e_3 = 1 \\ &\text{membership function variance}: \quad \beta_1 = 0.25, \beta_2 = 0.25, \beta_3 = 0.25 \\ &e_{1\,\min} = -8, e_{1\,\max} = 8, e_{2\,\min} = -5, e_{2\,\max} = 5, e_{3\,\min} = -4, e_{3\,\max} = 4 \\ &\beta_{1\,\min} = 0, \beta_{1\,\max} = 1, \beta_{2\,\min} = 0, \beta_{2\,\max} = 1, \beta_{3\,\min} = 0, \beta_{3\,\max} = 1 \\ &\left|e_{1threshold}\right| = 2, \left|e_{2threshold}\right| = 1.25, \left|e_{3threshold}\right| = 1 \end{aligned} \tag{42}$$

Phase diagram of state trajectory for the drive system with the nominal parameters is shown in Fig. 5.

The estimation errors of parameters a, b, c obtained from the hybrid control approach were near origin, as demonstrated in Fig. 6. Also, disturbance estimation error with the hybrid controller was close enough to the origin, as shown in Fig. 7.

The error dynamics of e_1, e_2, and e_3 for optimal control, active control, adaptive backstepping, and hybrid control method are shown in Figs. 8, 9, and 10, respectively.

The synchronization errors of hybrid controller had less variations and reached the steady state level (origin) faster than those of optimal control, active control and adaptive backstepping control as evidenced by Figs. 8–10.

The stability curve of the overall system with the hybrid controller along with the nominated storage function diagram are shown in Fig. 11. The supply rate is negative definite for $u_t(t) = 0$, $\gamma_1 = -1$ and $\gamma_2 = 1$(see Appendix A). The stability curve has negative values the entire time indicating the inequality of (39) satisfies ($Q = 0.5 \times I_{3\times3}$). According to Theorem 4.1, the storage function is monotone and non-increasing after a certain time ($t_0 = 9$ s) in the neighborhood of origin demonstrating the stability of closed-loop system.

To further investigate comparison studies between the synchronization methods, another set of parameters for the Rossler system was considered. The results for nominal values of $a = 0.1$, $b = 0.1$, $c = 14$, which provides a chaotic behavior for the Rossler system, are presented. Figs. 12–15 demonstrate the phase diagram and error dynamics e_1, e_2, e_3 for this new set of parameters.

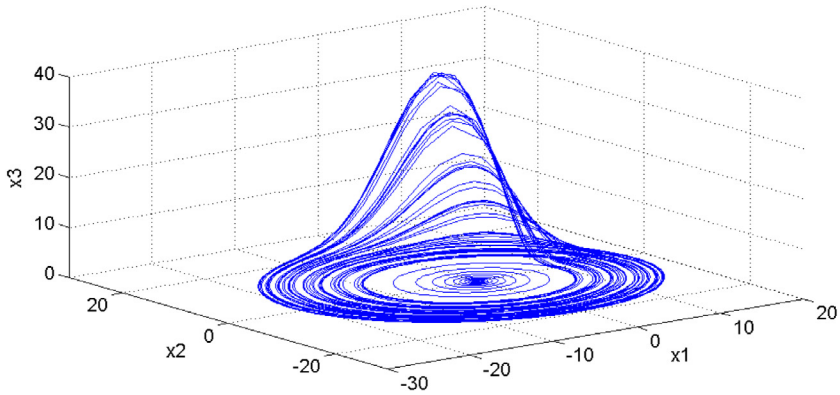

Fig. 5. Phase diagram of states trajectory for drive Rossler system.

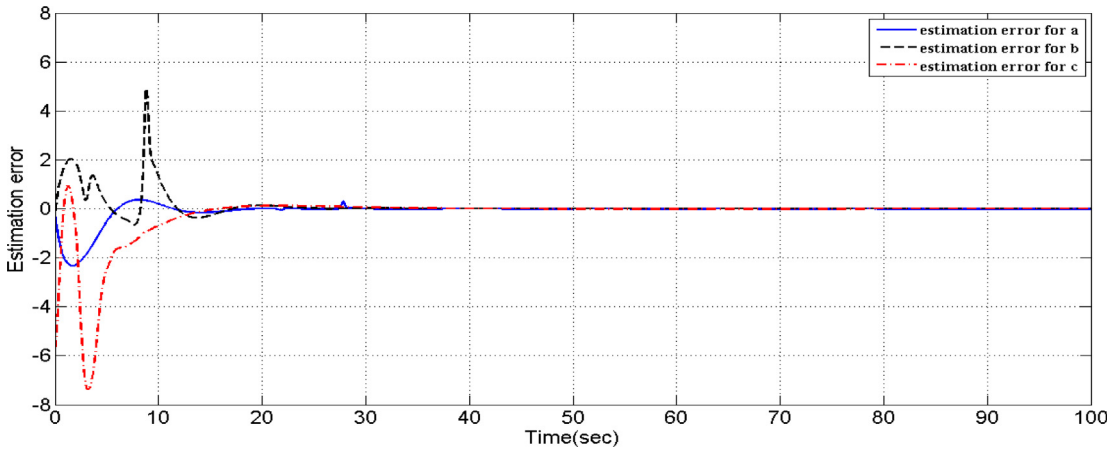

Fig. 6. Estimation errors for parameters a, b, c with hybrid control approach.

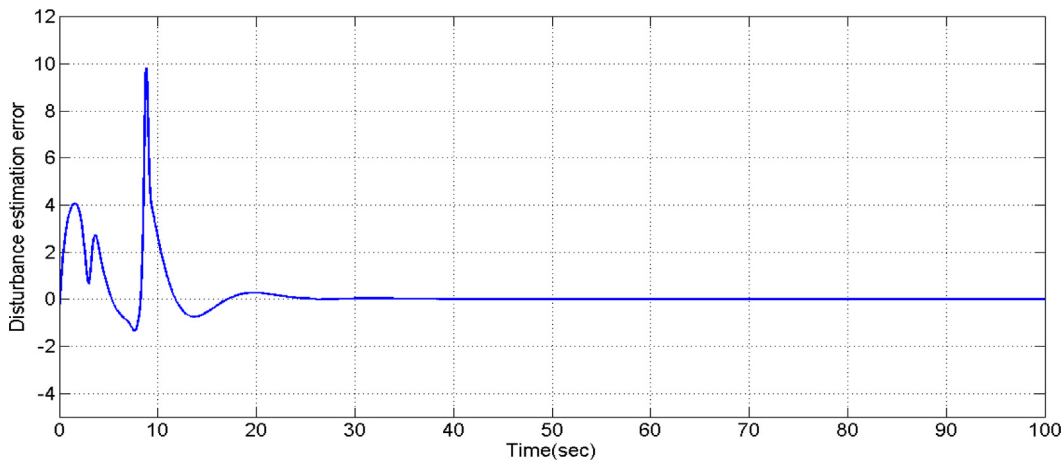

Fig. 7. Disturbance estimation error with hybrid control approach.

5.2. Performance analysis

The synchronization error dynamics e_1, e_2, and e_3 of the hybrid adaptive backstepping/sliding mode controller had smaller values and reached the steady state level (origin) faster than those of optimal control, active control and adaptive backstepping

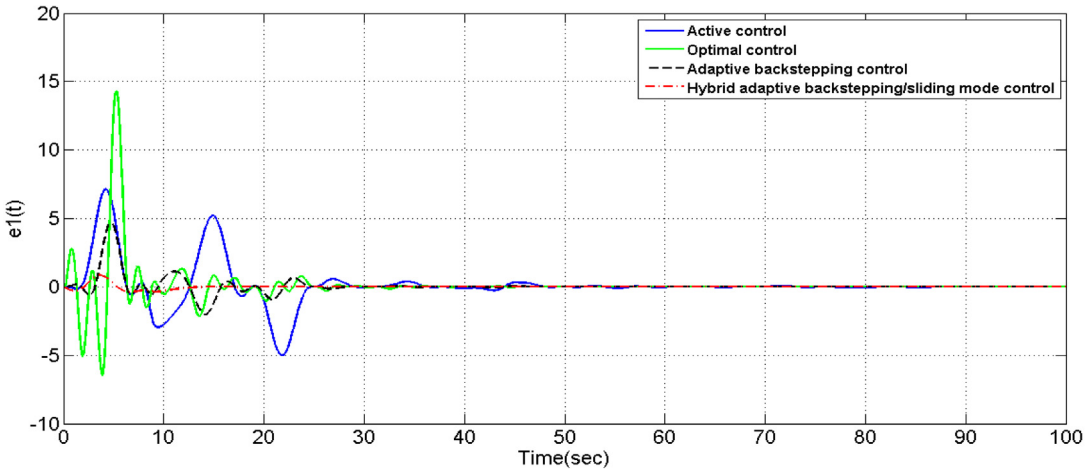

Fig. 8. Error dynamic of e_1 for active control, optimal control, adaptive backstepping control, and hybrid adaptive backstepping/sliding mode control.

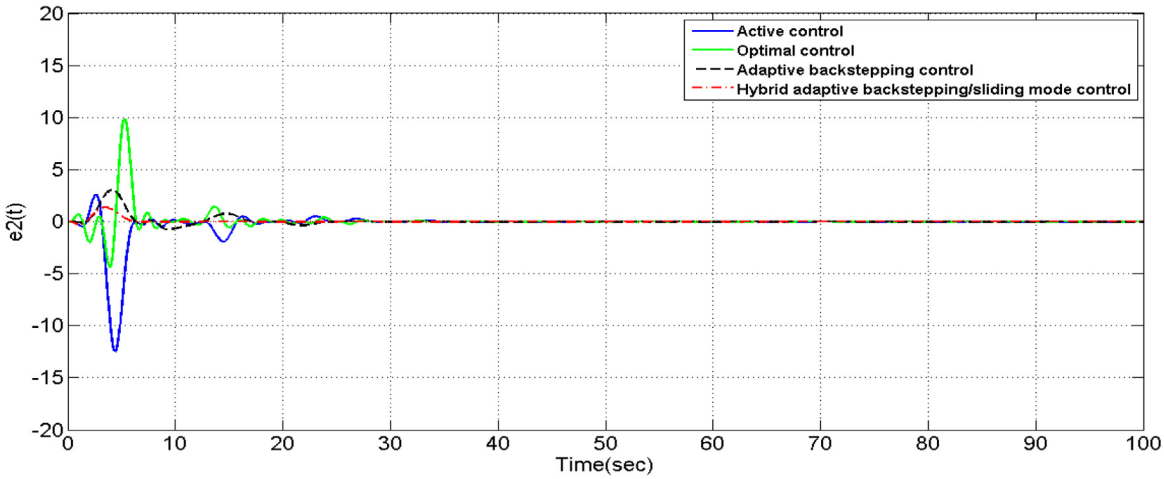

Fig. 9. Error dynamic of e_2 for active control, optimal control, adaptive backstepping control, and hybrid adaptive backstepping/sliding mode control.

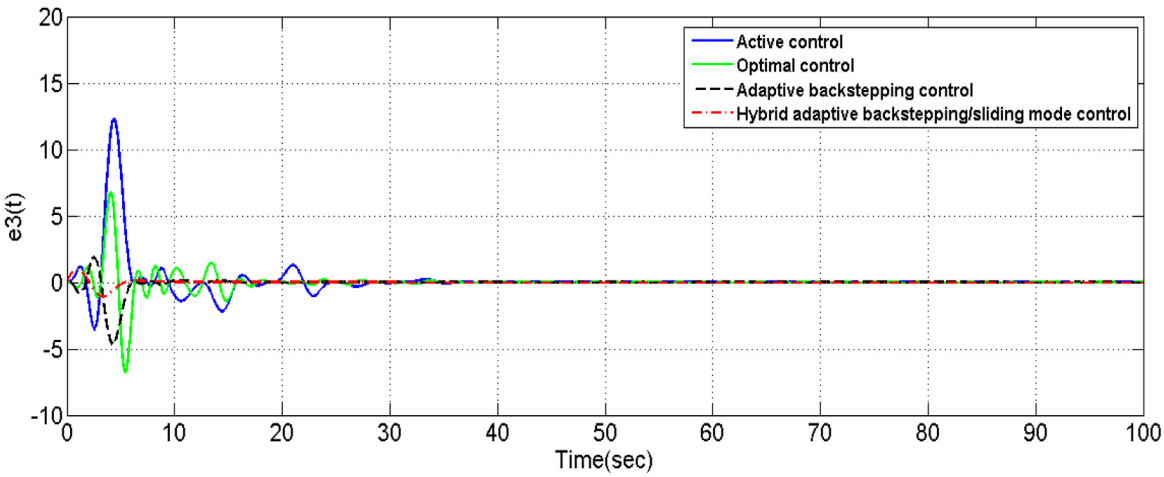

Fig. 10. Error dynamic of e_3 for active control, optimal control, adaptive backstepping control, and hybrid adaptive backstepping/sliding mode control($a = 0.2$, $b = 0.2$, $c = 5.7$)

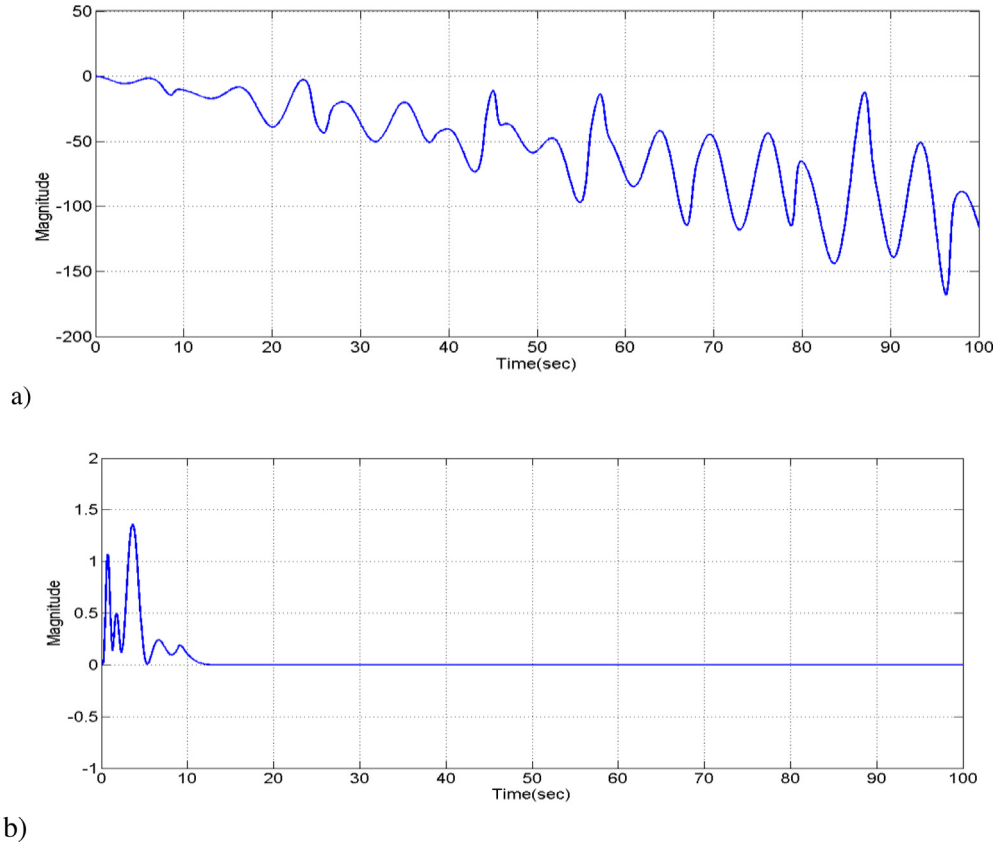

Fig. 11. (a) Stability substantiation diagram via dissipativity theory (for $\gamma_1 = -1$ and $\gamma_2 = 1$) (b) Nominated storage function diagram.

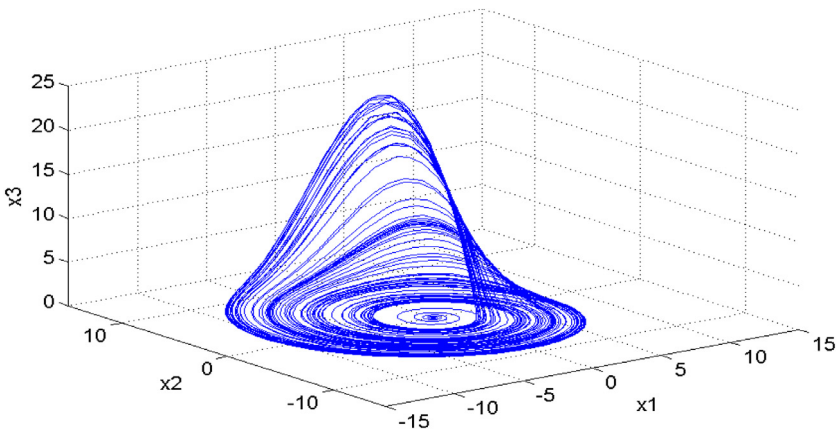

Fig. 12. Phase diagram of states trajectory for drive Rossler system with $a = 0.1$, $b = 0.1$, $c = 14$.

control, as evidenced by Figs. 8–10 and Figs. 13–15. To statistically compare the performance of the four controllers, one-way analysis of variance (ANOVA) with a significance level of 1% was deployed. The settling time (Ts), mean absolute error (MAE), and root-mean-square error (RMSE) of the synchronization errors e_1, e_2, and e_3 were considered as the performance measures. Ts, MAE, and RMSE of synchronization errors for the optimal control, active control, adaptive backstepping control, and hybrid control along with the p-values of ANOVA tests are shown in Table 3. The synchronization error e_1 of the hybrid controller reached the origin around 45% faster than the adaptive backstepping, 68% faster than the active control approach, and 34% faster than optimal control. Likewise, the synchronization errors e_2 and e_3 of the proposed hybrid controller settled to the vicinity of the origin much faster than those of the other three controllers (see the settling times in Table 3). Both MAE and RMSE of the synchronization errors of the hybrid controller were smaller than those of adaptive backstepping, active control, and optimal control approach. In fact, the

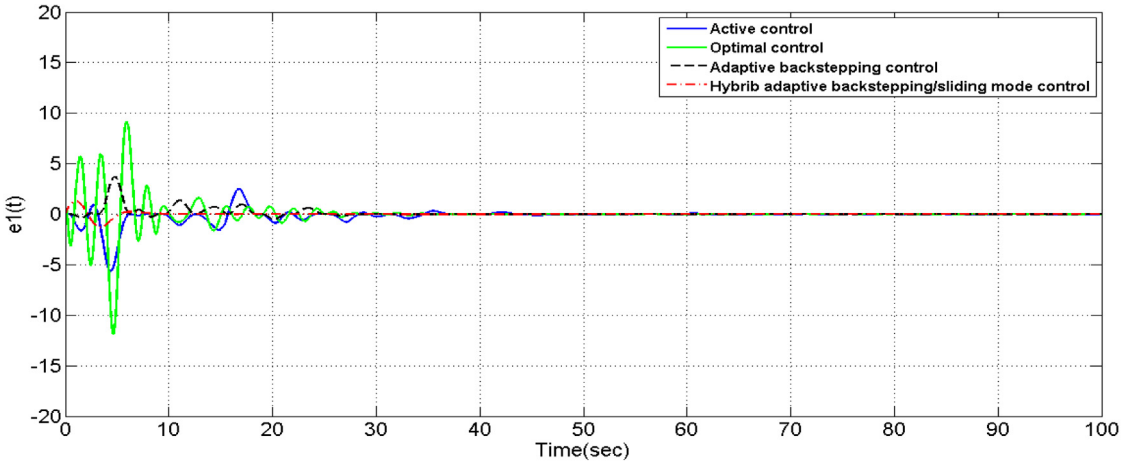

Fig. 13. Error dynamic of e_1 for active control, optimal control, adaptive backstepping control, and hybrid adaptive backstepping/sliding mode control ($a = 0.1$, $b = 0.1$, $c = 14$).

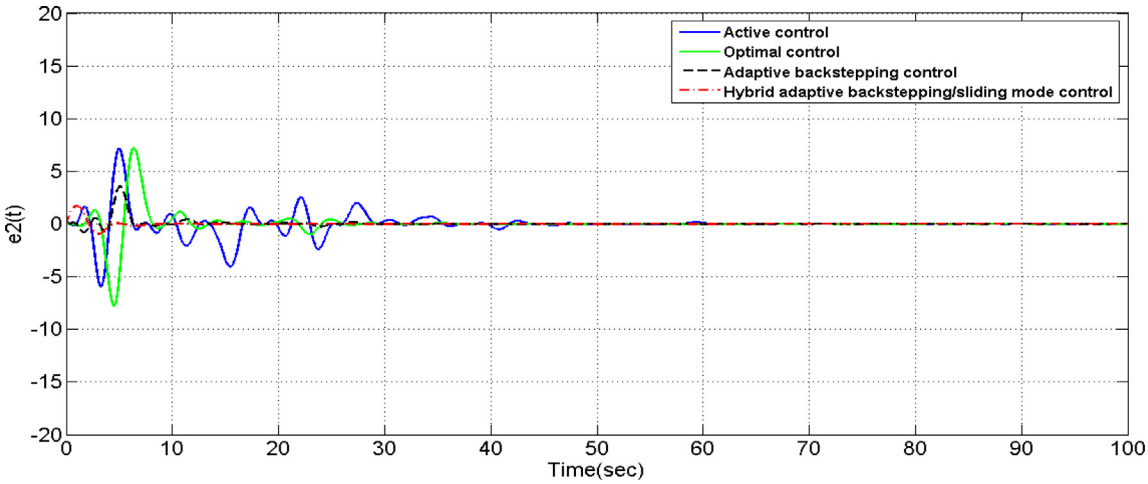

Fig. 14. Error dynamic of e_2 for active control, optimal control, adaptive backstepping control, and hybrid adaptive backstepping/sliding mode control ($a = 0.1$, $b = 0.1$, $c = 14$).

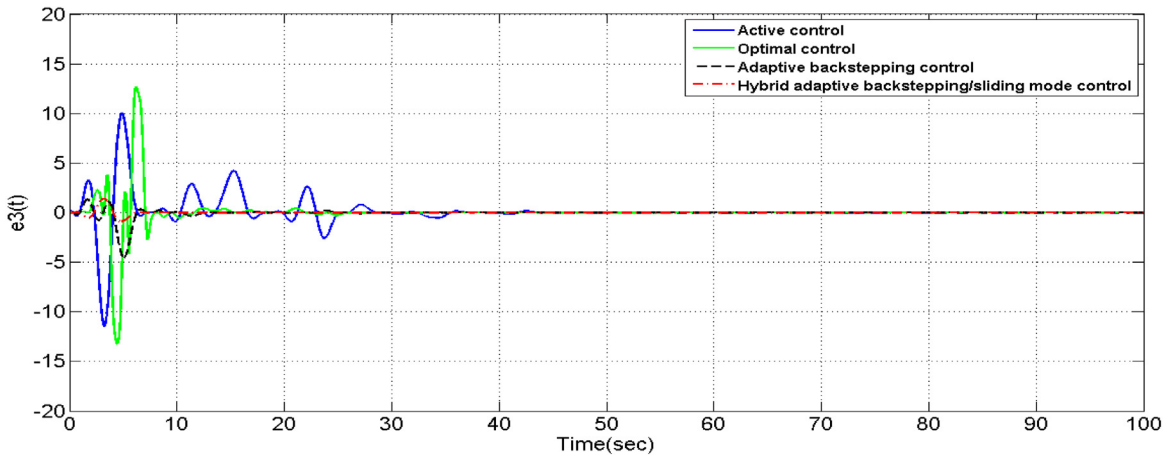

Fig. 15. Error dynamic of e_3 for active control, optimal control, adaptive backstepping control, and hybrid adaptive backstepping/sliding mode control ($a = 0.1$, $b = 0.1$, $c = 14$).

Table 3
Performance indexes of synchronization errors for the four control approaches: optimal control, active control, adaptive backstepping control, and hybrid control along with the p-values of ANOVA tests for the numerical example.

	Synchronization error	Optimal control	Active control	Adaptive backstepping	Hybrid control	ANOVA P-value
Settling time (s)	e_1	26.75	58.36	34.81	19.17	<0.01
	e_2	24.25	40.63	30.08	14.34	<0.01
	e_3	31.46	38.30	21.36	12.53	<0.01
Mean absolute error	e_1	0.39	0.58	0.19	0.04	<0.01
	e_2	0.22	0.28	0.12	0.03	<0.01
	e_3	0.23	0.35	0.10	0.03	<0.01
Root-mean-square error	e_1	1.30	1.47	0.65	0.15	<0.01
	e_2	0.98	1.38	0.46	0.18	<0.01
	e_3	0.85	1.40	0.53	0.17	<0.01

synchronization errors of hybrid controller had smaller values (higher accuracy) than the adaptive backstepping, which in turn had smaller errors than the active control and optimal control.

The ANOVA tests for all three performance measures had the p-values <0.01 implying that the statistically significant difference among the performance measures of the four controllers existed (i.e., at least one controller performance differed from other controller performances for each performance measure). Since a significant difference was detected by the ANOVA test, a post-hoc multiple comparison via Tukey test was conducted to determine what specific pair of controllers showed difference and what pairs did not.

The Tukey test results with 99% confidence interval suggested that the mean values of Ts, MAE, and RMSE of the synchronization errors from each control approach was notably different than those of the other three approaches (all p-values were less than 0.01). In particular, there were significant differences between the mean values of hybrid controller and adaptive backstepping, between the means of hybrid and active control, and between the mean values of hybrid and optimal control approach for the performance measures considered. These results clearly demonstrate the superior performance of the hybrid control method over the three state-of-the-art approaches, i.e. optimal control, active control, and adaptive backstepping control, for the synchronization of Rossler chaotic systems.

It should be noted that the Rossler system was selected as an example model due to the simplicity of its equations and the fact that Rossler systems were successfully utilized in secure communication and other application areas where accurate, quick synchronization was highly demanded. However, the proposed hybrid controller is not limited to the Rossler systems and can be applied to the other types of chaotic systems, as well. Furthermore, the proposed control approach is not limited to the identical models and can be very well used in synchronization between non-identical chaotic systems, where the response system model is different than the drive system model.

Another key aspect differentiating this work from others is the inclusion of unknown disturbances in chaos synchronization formulation and control law derivation. In a practical application like secure communication, there might exist external sources of uncertainties such as unknown disturbances that can potentially affect the performance of chaos synchronization systems. The proposed hybrid controller that combines adaptive backstepping approach with a fuzzy sliding mode control is able to successfully estimate such unknown disturbance, as shown in Fig. 7, and provide fast, high accuracy synchronization in presence of unknown system parameters and external disturbances.

6. Conclusion

In this paper, a hybrid adaptive backstepping/fuzzy sliding mode control was developed to provide fast, high accuracy synchronization of chaotic systems in presence of unknown system parameters and external disturbances. The superior performance of the proposed hybrid control approach was demonstrated against three state-of-the-art synchronization techniques, optimal control, active control, and pure adaptive backstepping, for a numerical example of Rossler system. The hybrid control method can be utilized in the application areas where the accelerated synchronization with high accuracy is demanded. The proposed synchronization approach is neither limited to Rossler systems nor to identical drive–response models. Synchronization performance of the proposed method for other types of chaotic systems including non-identical models can be investigated in future research studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Dissipativity theory [49]

Assume the dynamical system:

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)) \qquad f(0,0) = 0 \\ r(t) &= h(x(t), r(t)) \qquad h(0,0) = 0 \end{aligned} \tag{A.1}$$

where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the input, and $r(t) \in \mathbb{R}^p$ is the output. Also, $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ and $h : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^p$ are continuously differentiable mappings. The dissipativity theory characterizes a dynamical system with respect to the correlation of its inputs and outputs shown by a scalar-valued supply rate $J(u(t), r(t))$.

Definition A.1. The system (A.1) is dissipative with respect to a supply rate $J(u(t), r(t))$, if there exists $V : R^n \to R$ such that $V(0) = 0$, $V(x) \geq 0\ \forall x$, and

$$V(x(t)) - V(x(0)) \leq \int_0^\tau J(u(t), r(t))dt \tag{A.2}$$

for every input signal $u(.)$ and every $\tau \geq 0$ in the interval of existence of the solution $x(t)$. $V(.)$ is called a storage function.

Special Case: When $u(t) = 0$, a dissipative system whose supply rate $J(u, r)$ satisfies

$$J(0,0) = 0,\ J(0,r) \leq 0\ \ \forall r \in R^p \tag{A.3}$$

guarantees that trajectories starting in a sublevel set of ϑ remain in that sublevel set, because

$$V(x(\tau)) \leq \int_0^\tau \underbrace{J(0,r)}_{\leq 0} dt + V(x(0)) \leq V(x(0)) \tag{A.4}$$

Appendix B. Active control design

The derivation of active control law for the synchronization of Rossler chaotic systems is presented here. Consider a Rossler system as

$$\begin{aligned} \dot{x}_1 &= ax_1 + x_2 \\ \dot{x}_2 &= -x_1 - x_3 \\ \dot{x}_3 &= b + x_3(x_2 - c) \\ \\ \dot{y}_1 &= \hat{a}y_1 + y_2 + u_1 \\ \dot{y}_2 &= -y_1 - y_3 + u_2 \\ \dot{y}_3 &= \hat{b} + y_3(y_2 - \hat{c}) + u_3 + \hat{d} \end{aligned} \tag{B.1}$$

where x_i and y_i, $i = 1, 2, 3$ represent states dynamics for the drive and response system, respectively. Also, a, b, and c are constant known parameters, $\hat{a}$, $\hat{b}$, and $\hat{c}$ are unknown parameters, $\hat{d}$ is unknown external disturbance, and u_is, $i = 1, 2, 3$ are active control laws. In this approach, a separate control law for each state dynamics is needed. . The error dynamics system for active control approach is

$$\begin{aligned} \dot{e}_1 &= e_2 + \hat{a}y_1 - ax_1 + u_1 \\ \dot{e}_2 &= -e_1 - e_3 + u_2 \\ \dot{e}_3 &= (\hat{b} - b) + y_2y_3 - x_2x_3 - \hat{c}y_3 + cx_3 + u_3 + \hat{d} \\ (e_1 &= x_1 - y_1, e_2 = x_2 - y_2, \mathrm{e}_3 = x_3 - y_3) \end{aligned} \tag{B.2}$$

Active control laws are designed as

$$\begin{aligned} u_1 &= -(1 + \hat{a})e_1 \\ u_2 &= -e_2 - (x_3 - 1)e_3 \\ u_3 &= -(1 + y_2 - \hat{c})e_3 + \hat{d} \end{aligned} \tag{B.3}$$

and unknown parameters are estimated as

$$\begin{aligned}\dot{\hat{a}} &= -x_1 e_1\\ \dot{\hat{b}} &= -e_3\\ \dot{\hat{c}} &= x_3 e_3\\ \dot{\hat{d}} &= -2e_3\end{aligned} \tag{B.4}$$

Theorem B.1. *The system (B.2) is asymptotically stable by proposed control laws of (B.3) and parameter estimation of (B.4).*

Proof. The Lyapunov function is considered as

$$\begin{aligned}&\upsilon(t) = \frac{1}{2}(e_1^2 + e_2^2 + e_3^2 + e_a^2 + e_b^2 + e_c^2 + e_d^2)\\ &(e_a = \hat{a} - a, e_b = \hat{b} - b, e_c = \hat{c} - c, \mathrm{e}_d = \hat{d})\end{aligned} \tag{B.5}$$

The derivative of nominated Lyapunov function is

$$\begin{aligned}\dot{\upsilon}(t) &= e_1\dot{e}_1 + e_2\dot{e}_2 + e_3\dot{e}_3 + e_a\dot{e}_a + e_b\dot{e}_b + e_c\dot{e}_c + e_d\dot{e}_d\\ &= e_1[e_2 + \hat{a}y_1 - ax_1 + u_1] + e_2[-e_1 - e_3 + u_2] + e_3[(\hat{b} - b) + y_2y_3 - x_2x_3 - \hat{c}y_3 + cx_3 + u_3 + \hat{d}] + \cdots\\ &+\dot{e}_a(\hat{a} - a) + \dot{e}_b(\hat{b} - b) + \dot{e}_c(\hat{c} - c) + \dot{e}_d\hat{d}\\ &= e_1e_2 + \hat{a}y_1e_1 - ax_1e_1 + u_1e_1 - e_1e_2 - e_2e_3 + u_2e_2 + (\hat{b} - b)e_3 + y_2y_3e_3 - x_2x_3e_3 - \cdots\\ &-\hat{c}y_3e_3 + cx_3e_3 + u_3e_3 + \hat{d}e_3 + (\hat{a} - a)\dot{\hat{a}} + (\hat{b} - b)\dot{\hat{b}} + (\hat{c} - c)\dot{\hat{c}} + \hat{d}\dot{\hat{d}}\end{aligned} \tag{B.6}$$

Using the following equalities

$$\begin{aligned}&\hat{a}y_1e_1 - ax_1e_1 = \hat{a}{e_1}^2 + (\hat{a} - a)x_1e_1\\ &-\hat{c}y_3e_3 + cx_3e_3 = -\hat{c}{e_3}^2 - (\hat{c} - c)x_3e_3\\ &y_2y_3e_3 - x_2x_3e_3 = y_2{e_3}^2 + x_3e_2e_3\end{aligned} \tag{B.7}$$

the derivative of Lyapunov function (B.6) can be rewritten as

$$\begin{aligned}\dot{\upsilon} \;=\; & e_1e_2 + \hat{a}{e_1}^2 + (\hat{a} - a)x_1e_1 + u_1e_1 - e_1e_2 - e_2e_3 + u_2e_2 + (\hat{b} - b)e_3 + y_2{e_3}^2\\ &+\; x_3e_2e_3 - \hat{c}{e_3}^2 - (\hat{c} - c)x_3e_3 + u_3e_3 + \hat{d}e_3 + (\hat{a} - a)\dot{\hat{a}} + (\hat{b} - b)\dot{\hat{b}} + (\hat{c} - c)\dot{\hat{c}} + \hat{d}\dot{\hat{d}}\end{aligned} \tag{B.8}$$

Substituting (B.3) and (B.4) to (B.8) yields

$$\dot{\upsilon} = -{e_1}^2 - {e_2}^2 - {e_3}^2 \tag{B.9}$$

which is negative definite, so the error dynamic system of (B.2) is asymptotically stable in the equilibrium point of origin, according to the Lasalle theorem [42].

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